Regularization in High-Dimensions

QLS–MiCM Workshop

Why High-Dimensional Data Is Hard

Medical datasets, especially omics, often have:

- **Hundreds–thousands** of predictors.
- **Correlated** features.
- **Small sample sizes** relative to features.

Ordinary linear regression struggles with:

- unstable coefficients.
- overfitting.
- poor generalization.

This motivates regularization.

What Regularization Does

Regularization adds a **penalty** to large coefficients.

This encourages:

- smaller, more stable coefficients.
- less sensitivity to noise.
- better generalization on new data.

Two main types:

- **Ridge (L2):** shrinks all coefficients.
  - $$\hat{\beta} = \arg\min_{\beta} \left(\|y - X\beta\|_2^2 \; + \; \lambda \|\beta\|_2^2 \right)$$
- **Lasso (L1):** sets some coefficients to **zero**
  - $$\hat{\beta} = \arg\min_{\beta} \left(\|y - X\beta\|_2^2 \; + \; \lambda \|\beta\|_1 \right)$$

Ridge Regression (L2)

Ridge:

- handles correlated predictors well.
- spreads weight across related features.
- improves stability without eliminating variables.

Key question to explore:

Which features receive the most shrinkage?

Lasso Regression (L1)

Lasso:

- performs **automatic feature selection**.
- drives many coefficients exactly to **zero**.
- useful when features are redundant or weakly informative.

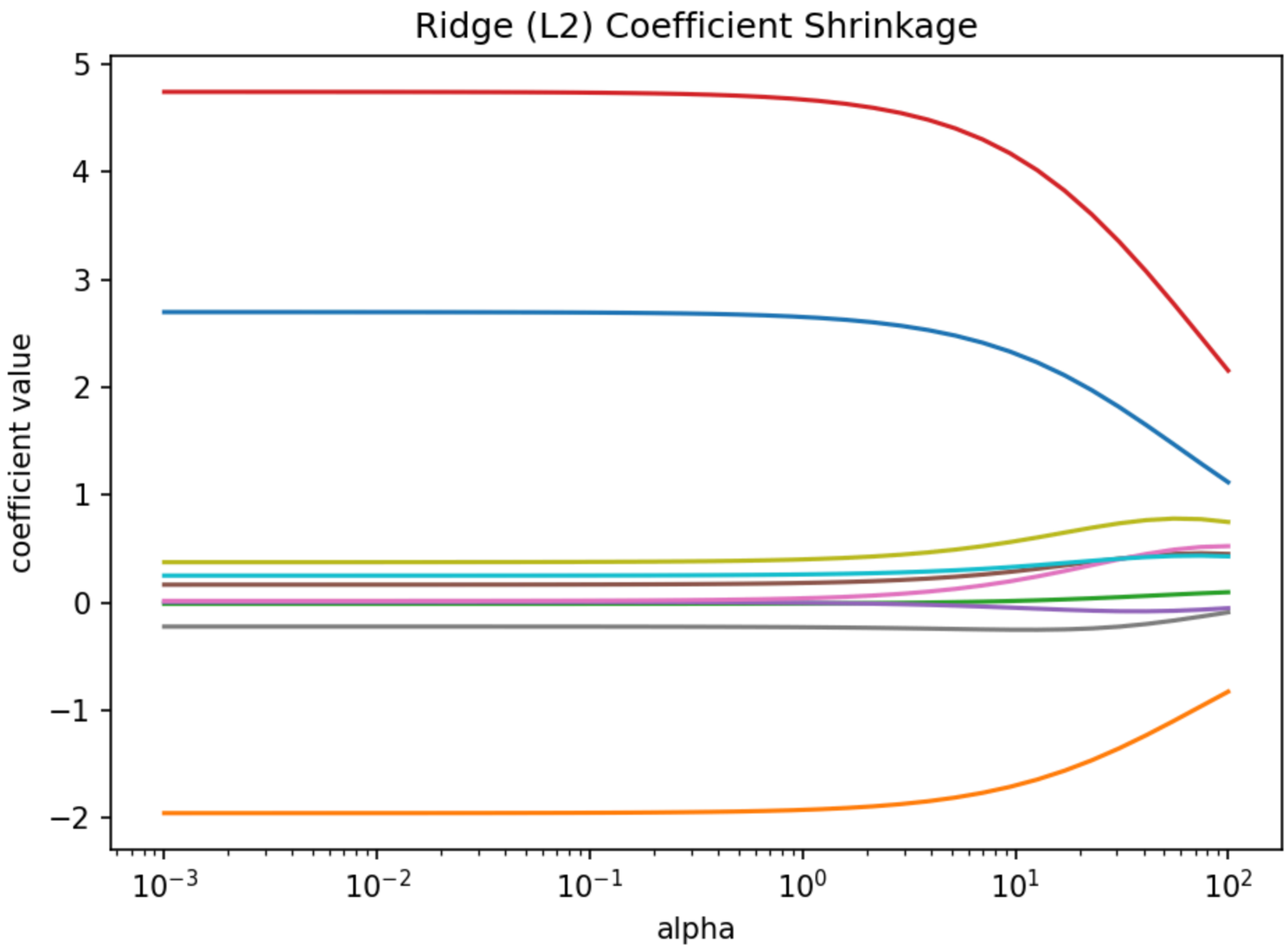
Key question to explore:

Which features does Lasso keep, and why?

Ridge (L2) Coefficient Shrinkage

As the regularization strength α **increases**:

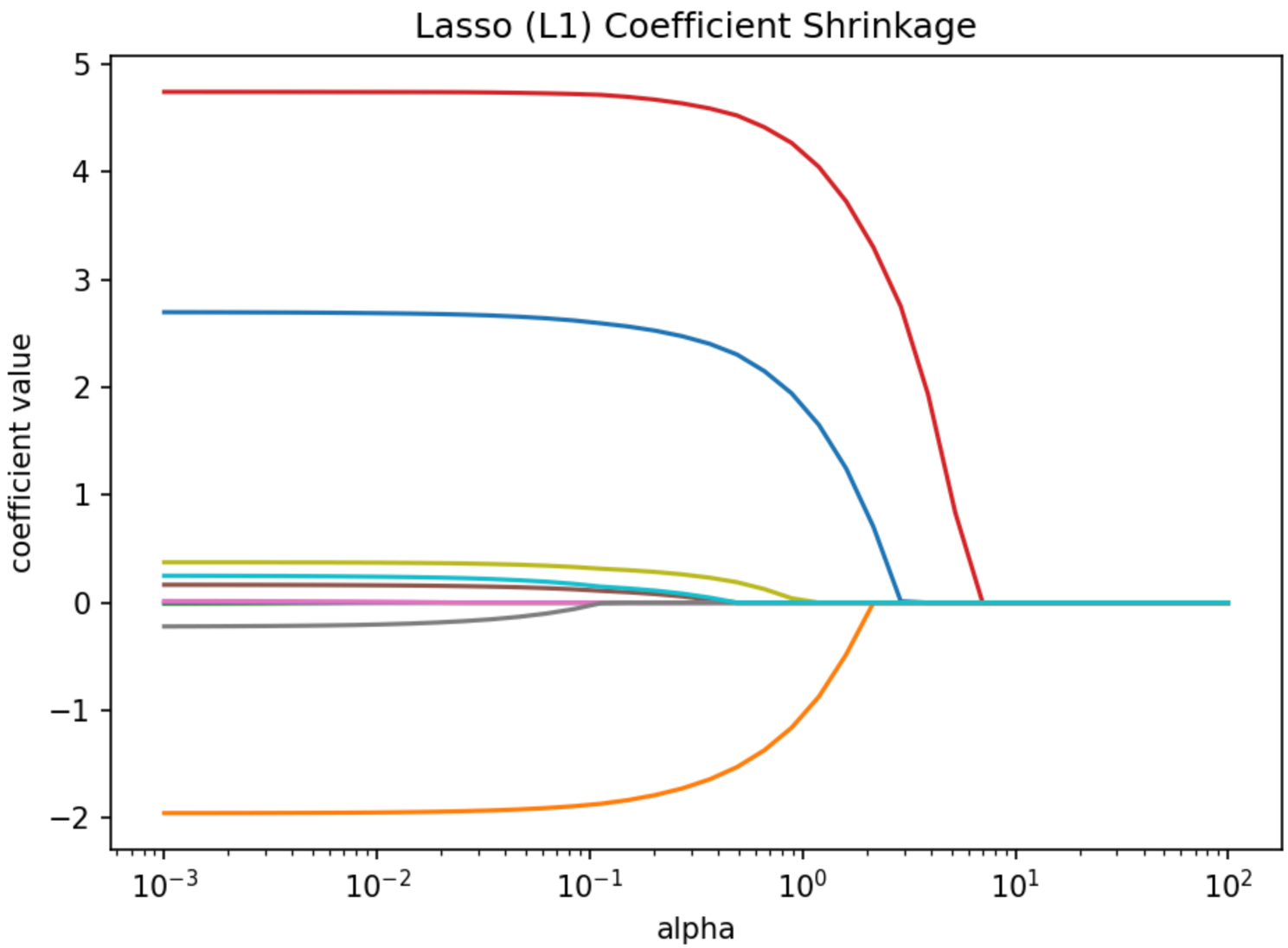
- All coefficients shrink **smoothly** toward zero.
- No coefficients drop to exactly zero.
- Ridge stabilizes the model but **keeps all features**.



Lasso (L1) Coefficient Shrinkage

As α increases:

- Coefficients shrink **unevenly**.
- Many coefficients drop to **exactly zero**.
- Lasso performs **automatic feature selection**.



Cross-Validation for α (Regularization Strength)

- Splitting your dataset into a training and validation set.
- Cross-validation uses more splits.
- We use cross-validation to estimate regularization strength.
- λ is the weight of the penalty (α is what scikit uses).